

Design and Analysis of an Electric Kart

by **Aman Suri**

1. Introduction to Electric Vehicles

An Electric Vehicle (EV) is a vehicle that uses an electric motor for propulsion, drawing energy from an onboard rechargeable battery pack rather than an internal combustion engine (ICE). EVs can be classified into three main types: Battery Electric Vehicles (BEVs), which rely entirely on electric power; Hybrid Electric Vehicles (HEVs), which combine an ICE with an electric motor; and Plug-in Hybrid Electric Vehicles (PHEVs), which feature a larger battery that can be charged externally. Key subsystems include the battery pack, power convertor, electric motor, and a vehicle control unit (VCU) that manages overall system operation [1].

1.1 Historical Background

Electric vehicles predate the ICE automobile. The earliest electric carriage was demonstrated by Robert Anderson around 1832, and by 1900 EVs accounted for roughly one-third of all vehicles on American roads. The mass production of Ford's Model T (1908) and the availability of cheap petroleum marginalised EVs for much of the twentieth century. Renewed interest emerged during the 1970s oil crisis, leading to U.S. government-funded R&D programmes. The modern era began with GM's EV1 (1996) and Toyota's Prius hybrid (1997), before Tesla's Roadster (2008) demonstrated that lithium-ion technology could deliver competitive range. By 2023, the global electric car stock exceeded 40 million units, driven by declining battery costs, government incentives, and tightening emissions legislation [3].

1.2 Advantages of Electric Vehicles

Electric vehicles offer several significant advantages over conventional ICE vehicles:

- **Zero emission:** EVs produce no pollutants at the point of use, improving urban air quality and reducing lifecycle CO₂ emissions by approximately 50% compared to petrol vehicles when powered from an average grid mix.

- **High energy efficiency:** Electric motors convert 85–95% of electrical energy into motion, compared to 20–35% for a petrol engine. Regenerative braking recovers a further 10–25% of kinetic energy [4].
- **Lower operating cost:** Electricity is cheaper per kilometre than petrol or diesel, and the reduced mechanical complexity of electric drivetrains results in approximately 40% lower maintenance costs over the vehicle lifetime.
- **Instant torque delivery:** Electric motors produce maximum torque from rest, providing immediate and responsive acceleration — a particularly valuable characteristic in kart and motorsport applications [1].
- **Reduced noise and vibration:** The absence of combustion events results in significantly lower noise, vibration, and harshness (NVH) levels, improving both driver comfort and environmental noise impact.

1.3 Examples of Electric Vehicles

A broad range of commercially available EVs illustrates the maturity of the technology. The Tesla Model S (2012) is a full-size BEV sedan with a range exceeding 600 km and sub-2.1-second 0–100 km/h acceleration. The Nissan Leaf (2010), one of the world's best-selling EVs with over 600,000 cumulative sales, demonstrated the viability of affordable mass-market BEVs. The Toyota Prius (1997) established the HEV segment and remains one of the highest-selling electrified vehicles globally. In the commercial sector, the Rivian R1T (2021) extended EV technology to full-size electric pick-up trucks.

2. Fundamentals of Electric Vehicles

2.1 Architecture of an Electric Vehicle

The architecture of an electric vehicle describes how its principal subsystems are interconnected to convert stored electrical energy into motion. Unlike an ICE vehicle, which relies on a mechanically coupled drivetrain, an EV architecture is largely electrical, enabling greater design flexibility and modularity. The four core subsystems and their functional relationships are illustrated in the block diagram below [4].

[**Energy Store**] → [**Power Electronics**] → [**Electric Motor**] →
[**Drivetrain / Wheels**]

Vehicle Control Unit (VCU)

The energy storage system (ESS) — typically a lithium-ion battery pack — supplies high-voltage DC power to the power electronics unit. The inverter within the power electronics block converts DC to AC (for AC motors) or regulates DC voltage for DC motors, controlling motor speed and torque in response to driver demand. The electric motor converts electrical energy into mechanical torque, which is transmitted through a single-speed reduction gearbox or direct drive to the wheels. A bidirectional link between the motor and battery enables regenerative braking, recovering kinetic energy during deceleration. The Vehicle Control Unit (VCU) acts as the supervisory controller, managing energy flow, safety interlocks, state-of-charge estimation, and thermal management across all subsystems [1][2].

Table 2.1 — Key EV Subsystems and Their Roles

Subsystem	Function	Typical Technology
Battery Pack	Energy storage and supply	Li-ion
Power Electronics	Voltage conversion and motor control	MOSFET inverter, DC-DC converter
Electric Motor	Convert electrical energy to torque	DC motor
VCU	System supervisory control	Embedded microcontroller / DSP
Regenerative Braking	Energy recovery during deceleration	Motor acting as generator

2.2 Battery — Energy Storage System

The battery pack is the primary energy storage component of a BEV and represents the single largest cost element, typically accounting for 30–40% of total vehicle cost. It stores electrical energy electrochemically and supplies high-voltage DC power to the drivetrain [1].

- **Function:** To store sufficient electrical energy to meet the vehicle's range requirements and deliver the peak power demanded during acceleration, whilst accepting charge from regenerative braking and external charging sources.
- **Types:** The dominant chemistry in modern EVs is lithium-ion (Li-ion), valued for its high energy density (150–250 Wh/kg), low self-discharge, and long cycle life. Variants include Lithium Iron Phosphate (LFP), which prioritises thermal stability and safety, and Nickel Manganese Cobalt (NMC), which offers higher energy density. Lead-acid batteries, whilst historically common, are limited to low-power auxiliary applications in modern EVs.
- **Operating principle:** During discharge, lithium ions migrate from the anode to the cathode through the electrolyte, releasing electrical energy. During charging, the process is reversed. Individual cells (~3.6 V) are arranged in series and parallel configurations to achieve the required pack voltage (typically 300–800 V) and capacity. A Battery Management System (BMS) monitors cell voltages, temperatures, and state of charge (SoC), providing protection against overcharge, over-discharge, and thermal runaway [4].

2.3 Power Converter

Power converters are the electronic interfaces between the battery, motor, and auxiliary systems. They regulate the magnitude, frequency, and direction of electrical power flow within the vehicle [5].

- **Function:** To convert and control electrical power between subsystems — specifically, to convert the battery's DC output into the appropriate form required by the motor, and to manage energy recovery during regenerative braking.
- **Types:** The primary converter is the DC/AC inverter, which transforms high-voltage DC from the battery into three-phase AC for AC motor drives. A DC-DC converter steps down the high-voltage bus (300–800 V) to the 12 V level required by auxiliary systems (lighting, control electronics). An on-board charger (OBC) converts mains AC to DC for battery charging. For DC motor drives, as used in this kart project, a DC-DC power converter (Chopper converter) regulates armature voltage directly.
- **Operating principle:** Inverters employ pulse-width modulation (PWM) of power semiconductor switches (IGBTs or MOSFETs) to synthesise a variable-frequency, variable-amplitude AC output from the DC bus. The switching frequency (typically 5–20 kHz) is controlled by the VCU based on driver torque demand. The duty cycle of the PWM

signal determines the effective output voltage and, consequently, motor speed and torque (Rashid, 2011).

2.4 Electric Motors and Generators

The electric motor is the primary actuator of the EV drivetrain, converting electrical energy into rotational mechanical energy. In regenerative braking mode, it operates as a generator, performing the reverse conversion.

- **Function:** To produce the torque and speed required to propel the vehicle, responding to driver demand via the inverter and VCU. In generator mode, to convert kinetic energy back into electrical energy for storage during deceleration.
- **Types:** Permanent Magnet Synchronous Motors (PMSMs) are the most widely used in modern EVs (e.g. Tesla Model 3) due to their high power density and efficiency. Brushless DC (BLDC) motors offer similar advantages with simpler control. Induction motors, used in earlier Tesla models, are robust and low-cost. DC Series-wound motors deliver very high starting torque and simple speed control via armature voltage variation, making them well-suited to low-cost educational and motorsport platforms.
- **Operating principle:** All electric motors operate on the principle of electromagnetic force: a current-carrying conductor in a magnetic field experiences a force ($F = BIL$). In DC motors, a commutator or electronic controller continuously switches current direction in the armature windings to sustain unidirectional rotation. The back-EMF ($\varepsilon = K\omega$) generated by the rotating armature opposes the supply voltage, providing an inherent self-regulating speed characteristic[7].

2.5 Regenerative Braking

Regenerative braking is the process by which kinetic energy, ordinarily dissipated as heat in conventional friction brakes during deceleration, is instead converted back into electrical energy and returned to the battery pack for subsequent use [8].

- **Function:** To improve overall vehicle energy efficiency by recovering a portion of the kinetic energy lost during deceleration. Studies indicate that regenerative braking can

improve energy efficiency by 10–25% depending on the drive cycle, with urban stop-start driving yielding the greatest benefit.

- **Types:** Regenerative braking is typically blended with conventional hydraulic friction braking. In a blended system, the control unit determines the optimal split between regenerative and friction braking torque based on battery SoC, vehicle speed, and deceleration demand. Some high-performance systems employ full hydraulic decoupling to maximise regenerative contribution.
- **Operating principle:** When the driver releases the throttle or applies the brake, the VCU commands the motor/inverter to operate in generator mode. The motor armature now acts as the primary source of braking torque, converting mechanical energy into electrical energy. This generated current is directed through the inverter back to the battery pack. The relationship between regenerative braking force and vehicle deceleration is governed by the motor's torque-speed characteristic and the inverter's current limiting capability. For a DC motor, the generated EMF must exceed the battery terminal voltage for current to flow back into the pack, requiring careful control of armature resistance or chopper duty cycle [4].

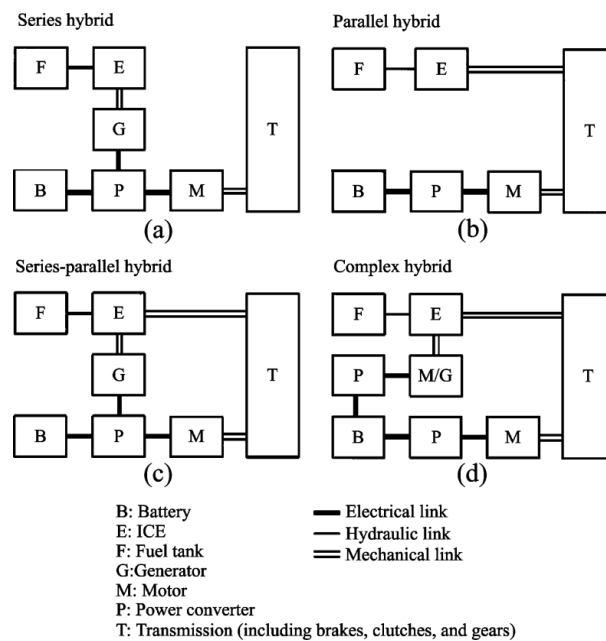


Figure 2.1 Four common architectures of HEV. [2]

3. Modelling and Simulation of a DC Motor-Driving System

3.1 Working Principle of a DC Motor

A DC motor is an electromechanical device that converts direct current electrical energy into rotational mechanical energy. It operates on the Lorentz force law: when a current-carrying conductor is placed in a magnetic field, it experiences a mechanical force:

$$F = B \cdot i \cdot L$$

where F is force (N), B is magnetic flux density (T), i is armature current (A), and L is the effective conductor length (m). In the motor, the armature windings carry the supply current within the field flux. The interaction between the armature current and the magnetic field produces torque, causing the rotor to rotate.

As the armature rotates, the conductors cut the magnetic flux and generate a back electromotive force (back-EMF) that opposes the supply voltage:

$$V_b = K_v \cdot \omega$$

where K_v is the back-EMF voltage constant (V·s/rad) and ω is the angular velocity (rad/s). The back-EMF reduces the net voltage driving current through the armature, thereby limiting current and speed at steady state. In an armature-controlled motor, the field flux is kept constant (by permanent magnets or fixed field current), and motor speed is controlled by varying the armature supply voltage V_a . This gives a linear input-output relationship, making the system amenable to Laplace-domain analysis[7].

Table 3.1 — Motor and Load Parameters (Mars Etec 48 V)

Symbol	Parameter	Value
R_a	Armature resistance	0.053 Ohm
L_a	Armature inductance	0.01 H
K_t	Torque constant	0.12 Nm/A
K_v	Back-EMF constant	0.127 V·s/rad

I	Load moment of inertia	2 kg · m ²
c	Rotary damping coefficient	8 Nm/ (rad/s)
V_a	Rated armature voltage	48 V

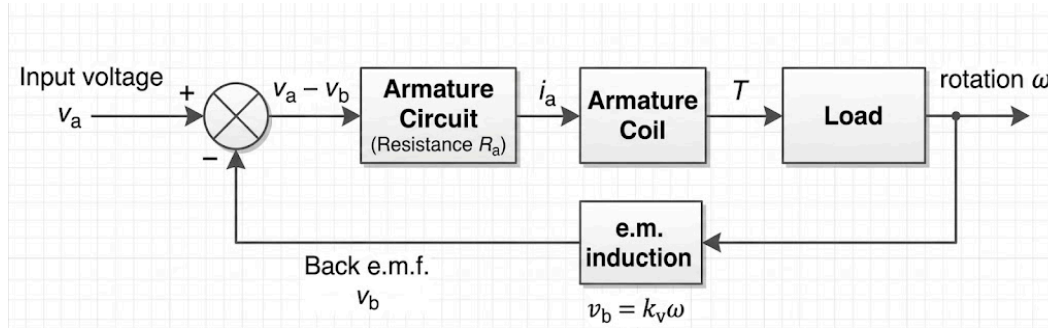


Figure 3.1 — Block diagram of the DC armature-controlled motor system [9]

3.2 Mathematical Models and Transfer Functions

The system is decomposed into three physical elements: the armature electrical circuit, the motor torque block, and the mechanical load. Each is modelled by its governing differential equation and Laplace-transformed to yield a transfer function block .

3.2.1 Armature Electrical Circuit — G1(s)

Applying Kirchhoff's Voltage Law around the armature loop:

$$V_a(t) = L_a \cdot \frac{di_a}{dt} + R_a \cdot I_a(t) + V_b(t)$$

Substituting $V_b(t) = K_v \cdot \omega(t)$ and applying the Laplace transform (zero initial conditions):

$$V_a(s) = (L_a \cdot s + R_a) \cdot I_a(s) + K_v \cdot \omega(s)$$

Rearranging gives the armature current as a function of the error signal $E(s) = V_a(s) - K_v \cdot \omega(s)$:

$$G1(s) = I_a(s)/E(s) = 1/(L_a \cdot s + R_a)$$

3.2.2 Motor Torque Generation — $G_2(s)$

The electromagnetic torque is proportional to armature current (constant field flux assumed):

$$T(t) = K_t \cdot I_a(t)$$

Laplace transform and transfer function:

$$G_2(s) = T(s)/I_a(s) = K_t = 0.12 \text{ Nm/A}$$

3.2.3 Mechanical Load — $G_3(s)$

Applying Newton's second law of rotation to the inertia-damper load:

$$T(t) = I \cdot \frac{d\omega}{dt} + c \cdot \omega(t)$$

Laplace transform and transfer function:

$$G_3(s) = \omega(s)/T(s) = 1/(I \cdot s + c)$$

3.2.4 Back-EMF Feedback — $H(s)$

The back-EMF feeds negatively back to the armature summing junction. Its transfer function is:

$$H(s) = K_v = 0.127 \text{ V} \cdot \text{s/rad}$$

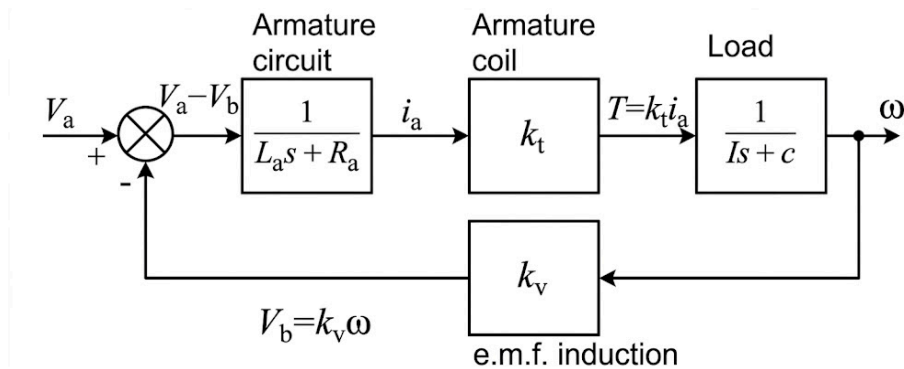


Figure 3.2 — Complete system block diagram with transfer functions [9]

3.3 Overall Transfer Function of the System

The three forward-path blocks in cascade give the open-loop transfer function:

$$G(s) = G_1(s) \cdot G_2(s) \cdot G_3(s) = K_t / [(L_a \cdot s + R_a)(I \cdot s + c)]$$

Applying the standard closed-loop formula with feedback $H(s) = K_v$:

$$\Omega(s)/V_a(s) = G(s) / [1 + G(s) \cdot H(s)]$$

Substituting $G(s)$ and $H(s)$:

$$\Omega(s)/V_a(s) = K_t / [(L_a \cdot s + R_a)(I \cdot s + c) + K_t \cdot K_v]$$

the closed-loop transfer function (second-order system) is therefore:

$$\Omega(s)/V_a(s) = 0.12 / (0.02 \cdot s^2 + 0.186 \cdot s + 0.43924)$$

This is a second-order transfer function with characteristic equation $0.02s^2 + 0.186s + 0.43924 = 0$. The system is stable (all coefficients positive). The steady-state speed gain ($s = 0$) is:

$$\Omega/V_a(s=0) = 0.12 / 0.43924 = 0.273 \text{ rad/s per Volt}$$

At the rated armature voltage of 48 V, the predicted steady-state angular velocity is:

$$\Omega_{ss} = 0.273 \times 48 = 13.1 \text{ rad/s } (\sim 125 \text{ rpm})$$

3.4 Simulation and Results

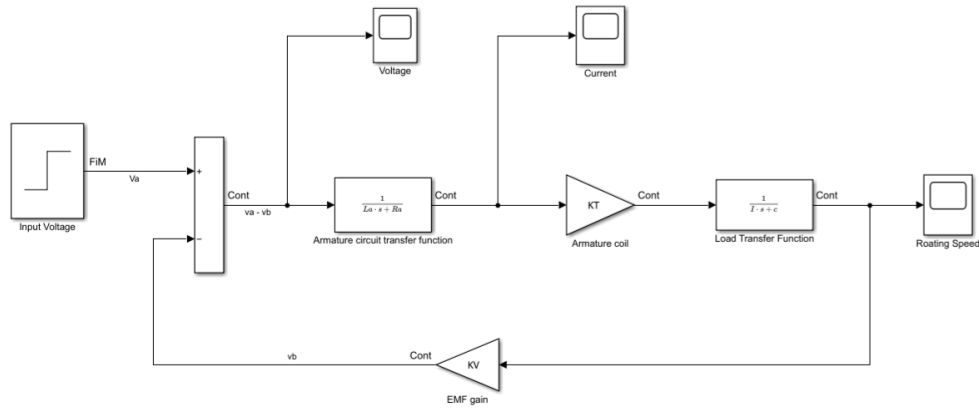


Figure 3.3 — SIMULINK Model of the DC Armature-Controlled Motor System

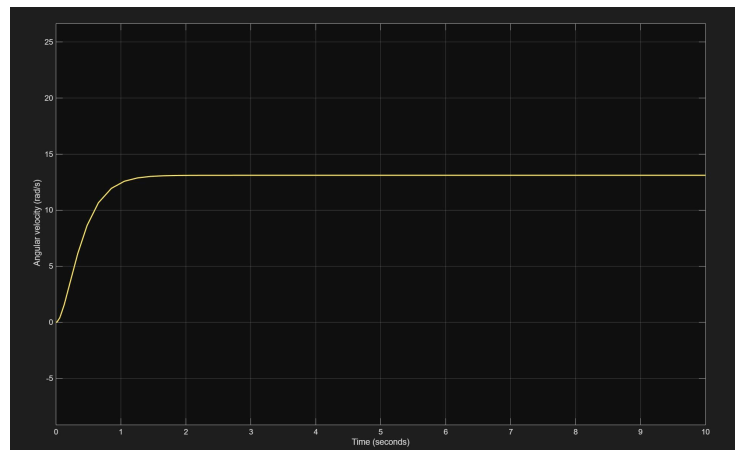


Figure 3.4 — Rotational Speed Response of the DC Motor to a 48 V Step Input Voltage (rad/s vs Time (s))

The SIMULINK model of the DC armature-controlled motor system was developed in accordance with the derived transfer function, incorporating the armature circuit block ($1/(L_a s + R_a)$), the torque gain block (K_T), the mechanical load block ($1/(J s + c)$), and the back-EMF negative feedback path (K). A step input of 48 V was applied to the armature, and the rotational speed response was recorded. As shown in Figure 3.4, the motor speed rises smoothly from rest, exhibiting a first-order exponential-like response with no overshoot, which is consistent with the overdamped nature of the second-order closed-loop system. The output reaches a steady-state

angular velocity of approximately 13.1 rad/s (≈ 125 rpm) at around $t = 2\text{--}3$ seconds, which is in close agreement with the analytically predicted steady-state value of $0.273 \times 48 = 13.1$ rad/s obtained from the closed-loop transfer function. This confirms that the mathematical model and the SIMULINK simulation are consistent, and that the Mars Etec 48 V DC motor, under the given load conditions, responds stably to a step voltage input without oscillation or instability.

4. Modelling and Simulation of an Electric Kart

This section describes the SIMULINK model used to simulate the behaviour of an electric kart navigating a lap circuit. The model replicates a real-world kart racing scenario by integrating six interacting subsystems: the driver model, power conversion, brake system, DC motor, vehicle dynamics, and battery. Together these subsystems simulate how the kart responds to a target speed profile derived from the track geometry, as shown in Figure 4.1.

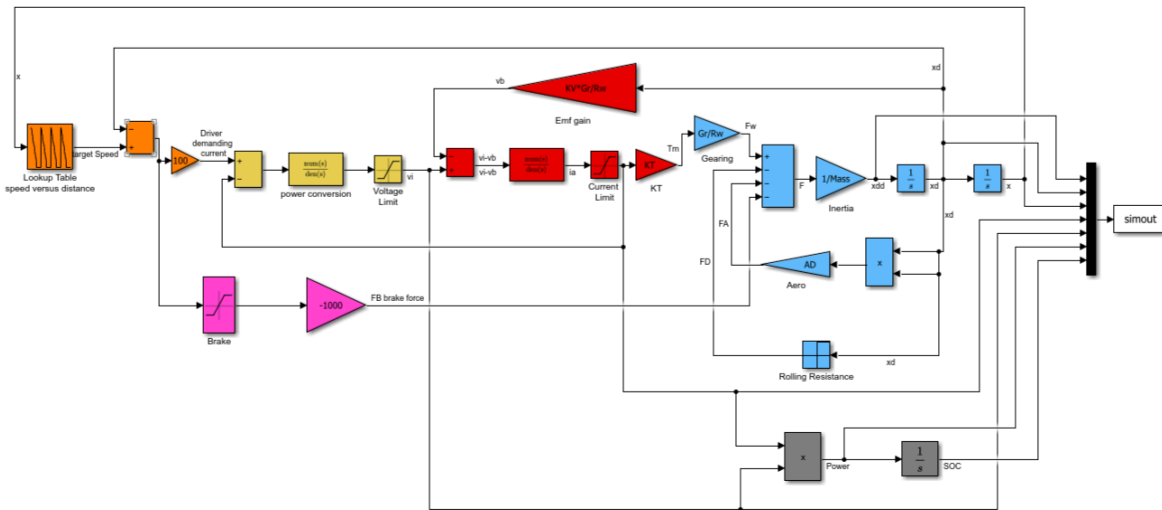


Figure 4.1 — SIMULINK Model of an Electric Kart

4.1 Target Speed Generation

Before the simulation runs, a target speed profile is pre-calculated from the geometry of the oval lap circuit and loaded into the model as a look-up table. The circuit has a total lap distance of 785 m, made up of two 100 m straights and two 180-degree bends, each with a radius of 25 m.

The maximum safe speed through each bend is set by the lateral acceleration limit. With a maximum lateral acceleration of 8 m/s^2 , the cornering speed is constrained to:

$$v_{\text{bend}} = \sqrt{a_{\text{lat}} \times r} = \sqrt{8 \times 25} = 14.14 \text{ m/s}$$

On the straights, the driver is assumed to accelerate as hard as possible from the bend exit up to the maximum achievable kart speed, then brake back down to 14.14 m/s before the next corner. The maximum braking deceleration is capped at 5 m/s^2 . Using these constraints, a target speed v_t is computed at positions x_t along the track using MATLAB and saved to $x_t.\text{mat}$. This data is loaded into the Lookup Table block in SIMULINK, which outputs the required target speed at any given track position during the simulation.

4.2 Model Elements

4.2.1 Driver Model

The driver model mimics how a racing driver responds to the track by continuously comparing the kart's current speed with the target speed at each position. It consists of two linked components. The Lookup Table block ('speed versus distance') takes the kart's current position (x_d) as input and returns the corresponding target speed (v_t) from the pre-loaded profile. A proportional Gain block then calculates the speed error — the difference between the target speed and the actual speed — and converts it into a demanded current signal. When the kart is slower than the target, a higher current is demanded to accelerate; when at or above target speed, the demand reduces or stops. This replicates how a real driver instinctively adjusts the throttle to match the required speed at every point on the circuit.

4.2.2 Power Conversion

The power conversion block represents the electronic control system that translates the driver's current demand into an actual voltage (v_i) supplied to the DC motor. It models the behaviour of the power electronics that regulate how much electrical power is delivered from the battery to the motor armature.

In the SIMULINK model, this is implemented as a first-order transfer function $(0.1s + 1)/(0.1s + 0.1)$, which introduces a small time delay representing the response lag of real power electronics. A Voltage Limit saturation block is placed at the output to ensure that the voltage supplied to the motor never exceeds the rated 48 V battery voltage.

4.2.3 Brake System

The brake system models the friction braking force applied when the kart needs to slow down before a corner. It runs in parallel with the motor drive path and acts independently of it. A Brake block detects when braking is required — that is, when the current kart speed exceeds the target speed for the upcoming bend. A Gain block of -1000 scales the brake signal into a large negative braking force (FB), which is fed directly into the vehicle dynamics subsystem to decelerate the kart. The maximum deceleration is limited to 5 m/s^2 in line with the track assumptions. This model uses friction braking only; regenerative braking is not included.

4.2.4 DC Motor

The DC motor subsystem is the same armature-controlled motor model developed in Section 3, now embedded within the broader kart simulation. It converts the voltage output from the power conversion block into a mechanical tractive force that propels the kart. The subsystem contains the following elements in sequence:

- **Summing junction (Add):** subtracts the back-EMF voltage (v_b) from the supply voltage (v_i) to give the net driving voltage ($v_i - v_b$).
- **Armature circuit TF — $1/(L_a s + R_a)$:** converts the net voltage into armature current (I_a), with $L_a = 0.01 \text{ H}$ and $R_a = 0.053 \text{ Ohm}$.
- **Current Limit block:** saturates I_a to prevent excessive current that could damage the motor.
- **Torque gain K_T :** converts armature current into motor torque ($T = K_T \times I_a$, with $K_T = 0.12 \text{ Nm/A}$).
- **Gearing block (G_r/R_r):** converts motor torque into a tractive force at the wheel (F_T), accounting for gear ratio $G_r = 2$ and wheel radius $R_r = 0.1 \text{ m}$.
- **Back-EMF feedback (K_V):** feeds the wheel speed back to the summing junction as $E_b = K_V \times \text{speed}$, closing the motor control loop ($K_V = 0.127 \text{ V} \cdot \text{s/rad}$).

4.2.5 Vehicle Dynamics

The vehicle dynamics subsystem models the physical motion of the kart by applying Newton's second law ($F = ma$) to calculate acceleration, velocity, and distance. Four forces are summed at the Add1 block to give the net force acting on the kart:

- **Tractive force (FT):** the forward driving force from the motor via the gearing.
- **Aerodynamic drag (FD):** a speed-dependent resistive force, $FD = 0.5 \times A \times Cd \times \rho \times v^2$, implemented via the Aero gain block and a Product block. With $A = 0.5 \text{ m}^2$, $Cd = 0.6$, and $\rho = 1.202 \text{ kg/m}^3$, the drag coefficient $AD = 0.1802$.
- **Rolling resistance (FR):** a constant resistive force $FR = \text{Mass} \times g \times RRc = 200 \times 9.81 \times 0.02 = 39.24 \text{ N}$, from the Rolling Resistance block.
- **Brake force (FB):** the friction braking force from the brake system, applied when decelerating for corners.

The net force is divided by the kart mass (1/Mass block, $\text{Mass} = 200 \text{ kg}$) to yield acceleration. This is integrated by Integrator to give velocity (xd , m/s), and integrated again by Integrator1 to give position (x , metres). The velocity is fed back to the driver model and to the motor back-EMF, closing the full simulation loop.

4.2.6 Battery

The battery subsystem tracks the energy consumed by the motor throughout the simulation. It provides a running measure of the state of charge (SOC) by integrating power consumption over time.

The supply voltage (v_i) and armature current (I_a) are multiplied together in the Product1 block to calculate instantaneous electrical power: $P = v_i \times I_a$ (in watts). This power signal is then fed into Integrator2, which integrates over the simulation time to give the total energy consumed in joules. This SOC value, along with acceleration, velocity, distance, current, voltage, and power, is output to the workspace via the simout block and plotted using the kartplot.m script across eight figures for full post-simulation analysis.

4.3 Back-EMF Voltage Equation Derivation

The back-EMF voltage (V_b) in the motor circuit is related to the vehicle velocity (x_d) through the mechanical transmission system. This section demonstrates why the following equation holds:

$$V_b = K_v \cdot G_r \cdot x_d / R_w$$

The back-EMF generated by a DC motor is directly proportional to its rotational speed. This is a fundamental property of DC machines, as described in Section 3.1:

$$V_b = K_v \cdot \omega_m$$

Where K_v is the back-EMF voltage constant (V·s/rad) and ω_m is the angular velocity of the motor shaft (rad/s).

The linear velocity of the kart (x_d) is related to the angular velocity of the wheel (ω_w) through the wheel rolling radius R_w . Assuming no wheel slip:

$$x_d = R_w \cdot \omega_w$$

Rearranging for wheel angular velocity:

$$\omega_w = x_d / R_w$$

Gear Ratio Relationship

The gear ratio G_r is defined as the ratio of motor angular velocity to wheel angular velocity:

$$G_r = \omega_m / \omega_w$$

This relationship arises from the conservation of power and line speed through the transmission. In a chain or gear drive, the tangential velocity at the point of contact is equal on both sides of the transmission, giving:

$$\omega_m \cdot R_m = \omega_w \cdot R_w$$

where R_m is the radius of the motor sprocket and R_w is the radius of the wheel sprocket. Rearranging:

$$G_r = \omega_m / \omega_w = R_w / R_m$$

For the electric kart, $G_r = 2$, meaning the motor shaft completes two full rotations for every one rotation of the wheel. From the gear ratio definition, the motor angular velocity is:

$$\omega_m = G_r \cdot \omega_w$$

Substituting the expression for ω_w :

$$\omega_m = G_r \cdot (x_d / R_w)$$

Substituting the expression for ω_m into the back-EMF equation:

$$V_b = K_v \cdot \omega_m = K_v \cdot G_r \cdot x_d / R_w$$

This confirms that the back-EMF in the motor circuit is directly proportional to the vehicle's linear velocity, scaled by the gear ratio and inversely proportional to the wheel radius. In the SIMULINK model, this relationship is implemented through the back-EMF feedback block (K_v), the gearing block (G_r / R_w), and the vehicle velocity output (x_d), which together close the feedback loop between the mechanical motion of the kart and the electrical behaviour of the motor circuit.

4.4 Simulation Results and Analysis

The baseline simulation with the given kartdata parameters produces a consistent and physically realistic set of results across all eight plots. The speed versus distance plot shows the kart closely following the target speed profile (red), accelerating on the straights to approximately 17–18 m/s and braking to 14.14 m/s through each bend. The acceleration plot shows sharp positive peaks ($\sim 3.5 \text{ m/s}^2$) during acceleration phases and sharp negative dips during braking, which repeats consistently each lap. The velocity plot confirms the kart reaches a steady repeating pattern of acceleration and braking over the 40 second simulation. Distance increases steadily and linearly overall, covering approximately 600 m. The current peaks at around 300 A during initial acceleration then settles, reflecting the high torque demand at low speed. Voltage is limited to 48 V during motoring and drops during braking. Power peaks at approximately 15,000 W during hard acceleration. The energy (SOC) plot shows a steady increase in energy consumed over time, reaching approximately 180,000 J by the end of the simulation, which is expected for continuous lapping.

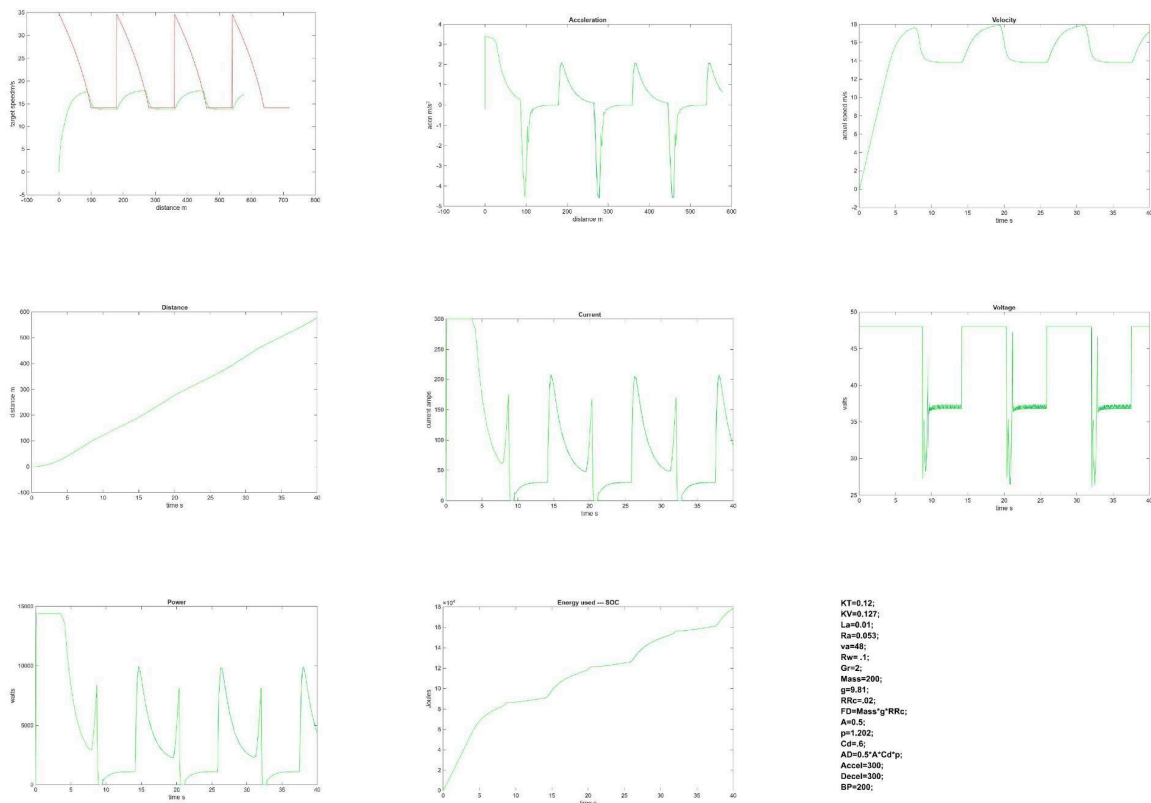


Figure 4.2 — Kart Simulation Results: Baseline Parameters (Mass = 200 kg, $R_w = 0.1 \text{ m}$)

4.4.2 Effect of Reducing Mass to 150 kg

Reducing the kart mass from 200 kg to 150 kg produces noticeable improvements in performance across several outputs. The speed versus distance plot shows the kart reaching a slightly higher peak speed of approximately 19–20 m/s on the straights compared to 17–18 m/s in the baseline, as the same motor force accelerates a lighter vehicle more quickly ($a = F/m$). The acceleration plot reflects this, showing higher peak acceleration values of approximately 4.5 m/s^2 compared to 3.5 m/s^2 in the baseline. The velocity plot confirms the kart builds speed faster after each corner. The distance plot shows the lighter kart covering slightly more ground in the same 40 seconds, indicating a faster lap. Current demand follows a similar pattern to baseline, though the lighter kart reaches target speed sooner so the high-current phases are shorter. Voltage behaviour remains comparable, still saturating at 48 V during acceleration. Power peaks are similar at $\sim 15,000 \text{ W}$ since the motor limits remain the same. Notably, the energy consumed is slightly lower than the baseline, as the kart spends less time at full current demand, making the lighter configuration both faster and more energy efficient.

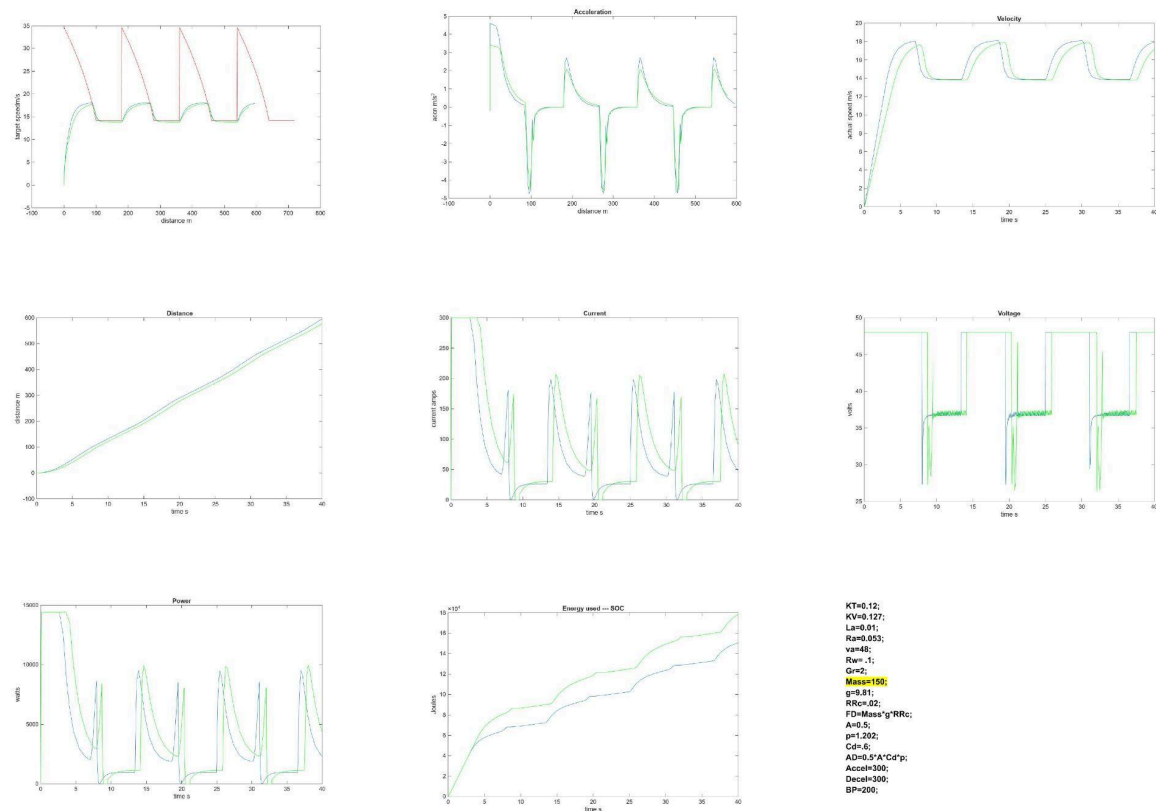


Figure 4.3 — Kart Simulation Results: Reduced Mass (Mass = 150 kg, $R_w = 0.1 \text{ m}$)

4.4.3 Effect of Increasing Wheel Radius to 0.125 m

Increasing the wheel radius from 0.1 m to 0.125 m has a more complex and mixed effect on performance. The speed versus distance plot shows the kart reaching a higher peak straight-line speed of approximately 20–22 m/s, as the larger wheel radius means each motor revolution covers more ground, effectively giving the kart a higher top speed. However, the acceleration plot shows reduced peak acceleration ($\sim 2 \text{ m/s}^2$) and also shows more negative deceleration spikes ($\sim -8 \text{ m/s}^2$), suggesting the kart struggles more to brake in time for corners. This is because the larger wheel reduces the effective gear ratio, meaning less torque is delivered to the road for the same motor output. The velocity plot shows a wider gap between the two simulation runs, with the new configuration reaching higher speeds but taking longer to accelerate from the bends. Current behaviour is more erratic, with higher sustained current draw as the motor works harder to maintain speed, and the voltage plot shows more variation with the voltage dropping lower during some phases. Power consumption is higher due to the increased current draw. Most significantly, the energy (SOC) plot shows considerably higher energy consumption — reaching approximately 250,000 J compared to 180,000 J in the baseline — meaning the larger wheel radius improves top speed at the cost of efficiency and driveability.

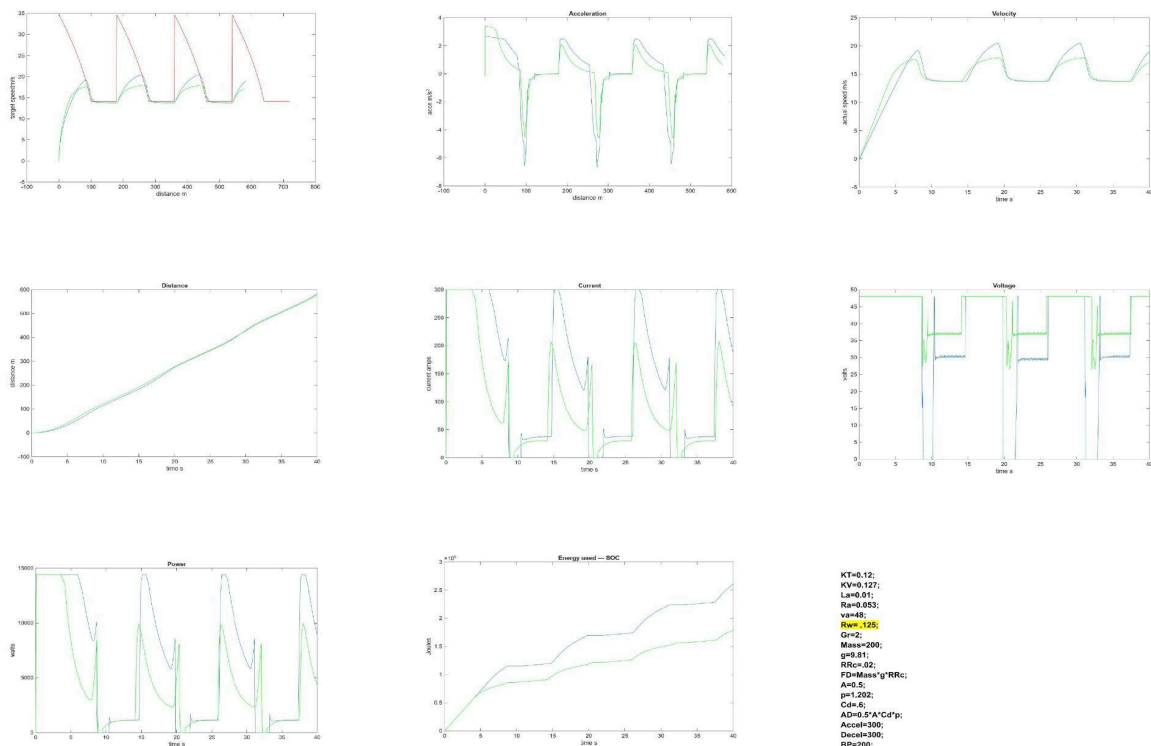


Figure 4.4 — Kart Simulation Results: Increased Wheel Radius (Mass = 200 kg, $R_w = 0.125 \text{ m}$)

5. Simulation of Kart Suspension System

5.1 System Overview

The suspension system which represents one part of the kart and captures the essential dynamics of wheel and body motion in response to road surface inputs. As shown in Figure 5.1, the model consists of three mechanical elements arranged vertically:

- **Sprung mass (m_s):** represents the vehicle body supported by the suspension
- **Suspension system:** consists of a spring (stiffness k_s) and a dashpot/damper (damping coefficient b_d) in parallel, connecting the sprung and unsprung masses
- **Unsprung mass (m_u):** represents the wheel and axle assembly, connected to the road via the tyre spring (stiffness k_T)

The road surface displacement x_R acts as the input to the system, and the sprung mass velocity v_s is the primary output of interest, representing how much the vehicle body moves in response to road disturbances.

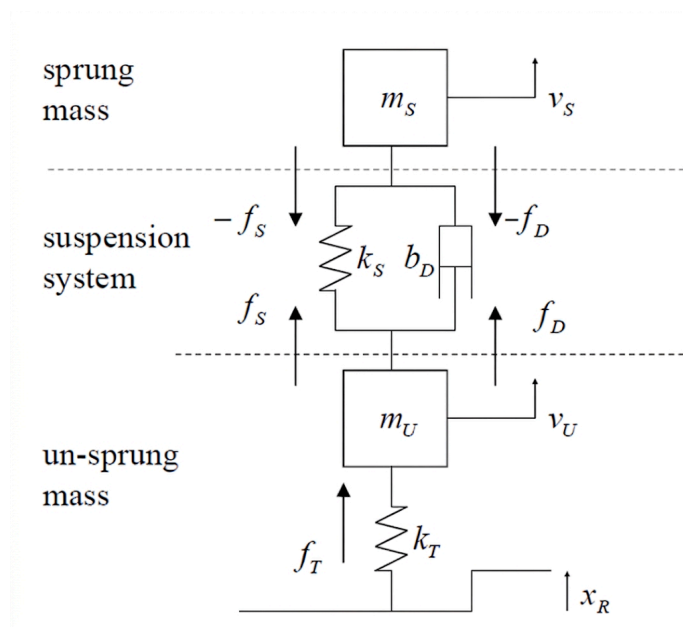


Figure 5.1 Physical Model of the Suspension System

5.2 Governing Equations

The system is described by five coupled equations,

The sprung mass equation:

$$m_s \cdot \frac{dv_s}{dt} = f_s + f_D$$

The spring force (suspension):

$$\frac{df_s}{dt} = k_s (v_u - v_s)$$

The damper force:

$$f_D = b_D (v_u - v_s)$$

The unsprung mass equation:

$$m_u \cdot \frac{dv_u}{dt} = f_T - f_s - f_D$$

The tyre force and wheel position:

$$f_T = k_T (x_R - x_U), \text{ and } \frac{dx_U}{dt} = v_U$$

These equations show that the suspension forces depend on the relative velocity and displacement between the sprung and unsprung masses, while the tyre force depends on the relative displacement between the road and the wheel.

5.3 SIMULINK Model

The full system is implemented in SIMULINK using three interconnected subsystems, as shown in Figure 5.2. A step input x_R represents the road disturbance (such as a curb or bump). This feeds into the unsprung mass subsystem, whose output (v_u) feeds into the suspension subsystem, whose output ($f = f_s + f_D$) feeds into the sprung mass subsystem, producing the final output

v_s . The feedback of v_s back into the suspension block closes the loop between the body and wheel motion.

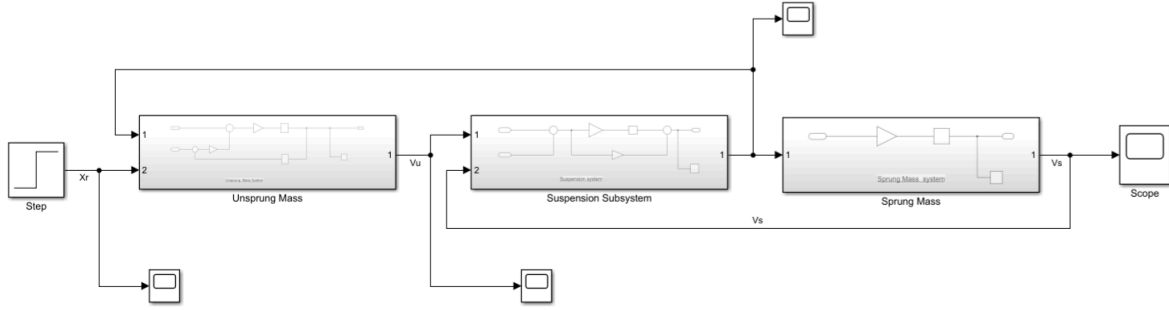


Figure 5.2 — SIMULINK Kart Suspension System Model (Top-Level Block Diagram)

Unsprung Mass Subsystem

As shown in Figure 5.3, this subsystem implements the equation $m_u \cdot \frac{dv_u}{dt} = f_T - f_s - f_D$. The tyre force $f_T = k_T (x_R - x_U)$ is computed by finding the displacement difference between road input and wheel position, multiplied by gain k_T . This is added to the suspension force input $f (= f_s + f_D)$ at the summing junction. The net force is scaled by $1/m_u$ to give acceleration a_u , integrated once to give wheel velocity v_u , and integrated again to give wheel position x_u , which feeds back to the tyre force calculation.

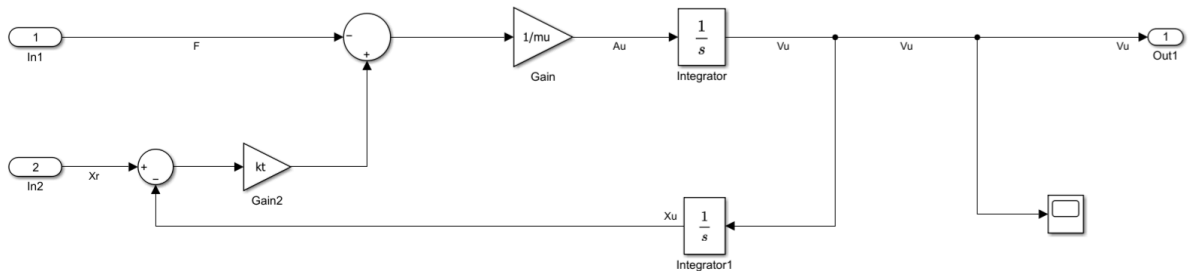


Figure 5.3 — SIMULINK Unsprung Mass Sub-System Model

Suspension Subsystem

Figure 5.4 shows the suspension subsystem implementing the spring and damper equations. The velocity difference ($v_U - v_S$) is computed at the summing junction. This is passed through the spring gain k_s and integrated to produce the spring force f_s . In parallel, the same velocity difference is passed through the damper gain b_D to produce the damping force f_D directly. The two forces are summed to give the total suspension force $f = f_s + f_D$, which is output to both the sprung and unsprung mass subsystems.

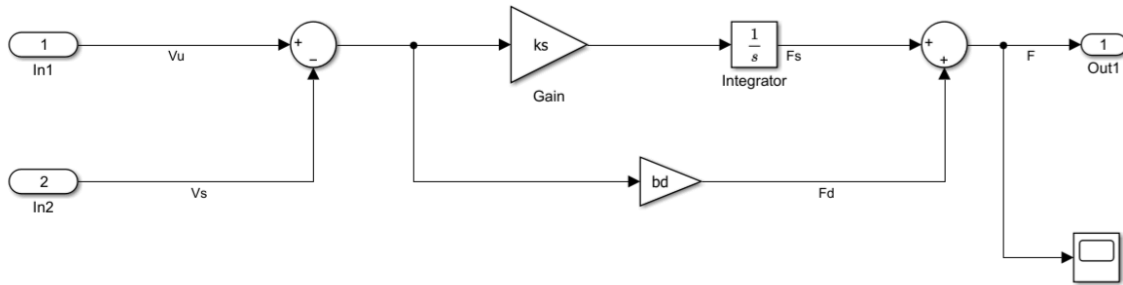


Figure 5.4 — SIMULINK Suspension Sub-System Model

Sprung Mass Subsystem

Figure 5.5 shows the simplest subsystem, implementing $m_s \cdot \frac{dv_s}{dt} = f_s + f_D$. The total suspension force f is scaled by the gain $1/m_s$ to give sprung mass acceleration, which is then integrated to produce the sprung mass velocity v_s . This is fed back to the suspension subsystem and output as the system response.

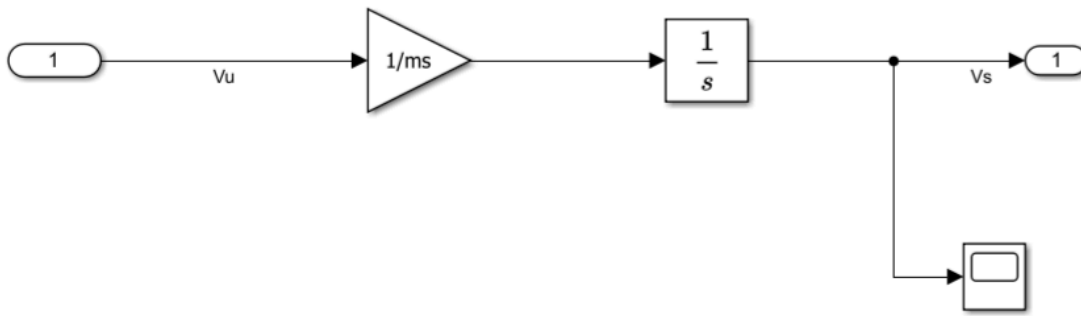


Figure 5.5 — SIMULINK Sprung Mass Sub-System Model

5.4 Simulation Results

Figure 5.6 shows the sprung mass velocity response v_s to a step road input x_R , representing the vehicle body's reaction to hitting a sudden bump.

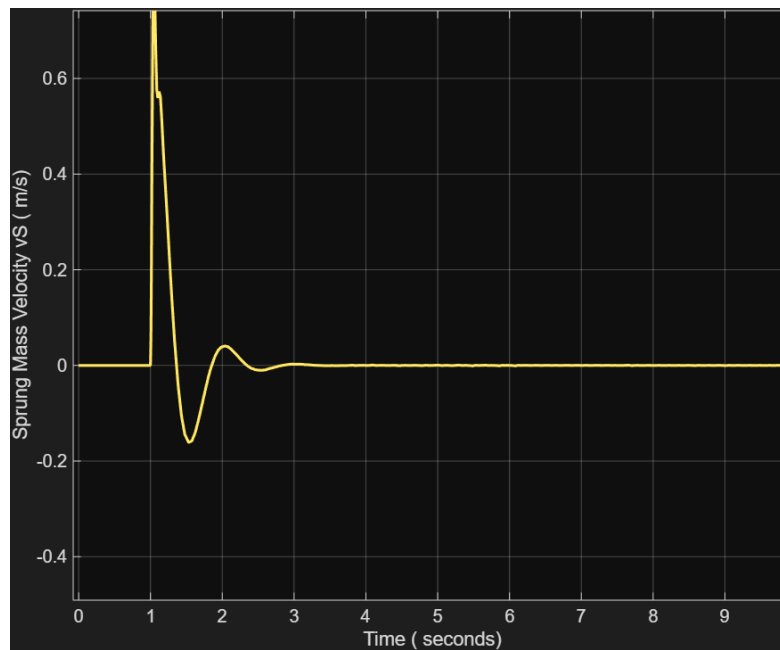


Figure 5.6 — Sprung Mass Velocity (v_s) Response to Step Road Input(x_R)

The sprung mass velocity rises sharply to a peak of approximately 0.82 m/s at around $t = 1$ second, immediately after the road disturbance is applied. The system then oscillates with

diminishing amplitude — the velocity drops to approximately -0.25 m/s before recovering — indicating an underdamped second-order response. By approximately $t = 4\text{--}5$ seconds, the oscillations have fully decayed and the velocity returns to zero, meaning the vehicle body has settled back to its equilibrium position. This damped oscillatory behaviour is physically expected: the spring stores and releases energy whilst the dashpot dissipates it, gradually reducing the amplitude of each oscillation until the system reaches steady state. The relatively quick settling time suggests adequate damping in the suspension system, which would provide acceptable ride comfort in a real vehicle application.

6. Conclusion

This coursework has been a great introduction to the world of automotive electronics and has helped me understand how many different engineering concepts come together in a real system like an electric vehicle.

Through the electric kart project, I was able to apply what I learned about control systems and see how mathematical models — which can seem quite abstract in theory — actually describe the real physical behaviour of a motor and a vehicle. Deriving the transfer function of the DC motor and then seeing the simulation produce a result that matched the analytical prediction was a particularly satisfying moment, as it showed that the theory genuinely works in practice.

Using MATLAB and SIMULINK throughout the course has been very useful. I have gone from being relatively unfamiliar with these tools to being able to build multi-subsystem simulation models, tune parameters, and interpret results with reasonable confidence. The kart lap simulation in particular showed me how individual subsystems — the driver model, motor, braking, and vehicle dynamics — all interact with each other, and how changing a single parameter like mass or wheel radius can affect the overall system behaviour.

The broader topics covered in the course were equally interesting. Learning about vehicle networks such as CAN bus helped me appreciate how modern cars manage communication between dozens of electronic control units. Data acquisition gave me an understanding of how real sensor data is collected and used. Studying ABS and ADAS systems showed me how the control theory we learned is applied to solve real safety-critical problems in vehicles that are on the road today.

Overall, this course has given me a much better understanding of how modern vehicles work at an electronic and systems level, and has equipped me with practical simulation skills that I feel will be genuinely useful going forward in my studies and career.

Finally, I would like to express my sincere gratitude to **Professor Yingping Huang** for the guidance and support provided throughout this course.

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- [9] Generated using Gemini AI
- [10] A significant portion of this report is derived from lectures conducted by Professor Yingping Huang.